

Parkinson's Tremor Monitoring & Alert System

1. Abstract

This report presents the design and implementation of a wearable Parkinson's Tremor Monitoring and Alert System built around a Seeed Studio XIAO ESP32-C3 microcontroller and an MPU-6050 six-axis inertial sensor. The system operates in two distinct phases: a one-week Calibration Phase that builds a patient-specific movement baseline through user-confirmed tremor labelling, followed by an Inference Phase in which a trained TCN classifier identifies pathological tremor events in real time and displays them on an interactive dashboard. Three clinically relevant test cases are demonstrated: thumb-tapping tremor, driving-related hand vibration, and consolidated random background motion. Post-detection, the system prompts the patient to confirm or deny each detected event via a binary Yes/No interface, continuously refining the personal model without requiring clinical intervention. The entire hardware bill of materials costs under ₹900, making continuous Parkinson's monitoring accessible outside hospital settings for the first time.

2. Literature Survey

Year	Paper Title / System	Key Contribution	Limitation
2023	PKG Watch — Global Kinetics Corp.	FDA-cleared wrist-worn Parkinson's monitor; proprietary algorithm for bradykinesia scoring.	Cost > ₹1.5 lakh; no user-confirmed labelling; population model.
2022	Sensor-Based Tremor Analysis Using IMU — IEEE Trans. Neural Syst.	6-axis IMU classification of Parkinson's tremor vs. essential tremor at 91% accuracy.	Requires clinical-grade ADC; not tested on ESP32-class hardware.
2021	Real-Time Tremor Detection with MPU6050 — Sensors MDPI	Low-cost MPU6050 wearable achieving real-time tremor flag; rule-based threshold.	No person-specific baseline; no severity scoring; no data labelling loop.
2021	Wearable IMU + Random Forest for PD Monitoring — J. Med. Eng.	Random Forest on time-domain IMU features; 87% sensitivity for tremor detection.	Trained on fixed population dataset; no adaptive calibration.
2020	Apple Watch Tremor Detection — npj Parkinson's Disease	Consumer smartwatch detects medication on/off states via accelerometry.	Proprietary; no data export; no caregiver alert system; cost > ₹40,000.
2019	UPDRS-Based Clinical Assessment — Mov. Disord. Soc.	Gold standard severity scale; validated by neurologists worldwide.	Clinic-only; 10-minute snapshot; subjective inter-rater variance up to 30%.

3. Problem Statement

Parkinson's disease is the fastest-growing neurological condition globally, with cases projected to double from 10 million in 2019 to nearly 20 million by 2050. India is expected to become the second-highest burden country within 5 years (Economic Times, 2026). Despite this scale, current clinical monitoring is fundamentally episodic:

Patients attend neurology clinics every 3–6 months for a UPDRS assessment lasting 10–30 minutes.

Tremor severity varies significantly by medication timing (on/off cycles), time of day, and emotional state - variations invisible to clinic snapshots.

No affordable, continuous system exists that can distinguish pathological tremors from everyday hand motion, generate validated labeled training data, or alert caregivers in real time. The result is missed early signals, inconsistent longitudinal records, and delayed medication adjustments , at the direct cost of patient independence.

The Gap in Current Care can be summarized as:

Existing commercial monitoring products such as the Kinesia ONE and PKG Watch cost thousands of dollars, require specialist setup, and output population-level severity scores that are insensitive to the unique movement signature of each individual patient. Apple Watch fall detection addresses falls, not Parkinson's-specific tremor characterisation. Academic research datasets (PhysioNet, mPower) rely on scheduled recording sessions, not continuous passive monitoring.

No currently deployed system combines: (a) continuous wrist-worn sensing, (b) user-confirmed data labelling in real time, (c) a personal baseline built from the individual's own movement history, and (d) a TCN classifier trained exclusively on that person's data. This work addresses all four simultaneously.

Dimension	Current Clinical Method	This System
Monitoring frequency	Every 3–6 months	Continuous, 24/7
Severity assessment	Subjective UPDRS by clinician	Objective ML severity score
Data quality	Clinician observation (10 min)	Validated user-confirmed labels
Personalization	Population-level norms	Person-specific baseline
Alert latency	Next clinic visit	Real-time caregiver notification
Cost	₹500–₹5000 per visit	₹800 one-time hardware

4. System Architecture

4.1 Hardware Components

Component	Specification	Role	Cost (₹)
Seed Studio XIAO ESP32- C3	Dual-core 32-bit RISC-V at 160 MHz; Wi-Fi 802.11 b/g/n; BLE 5.0; 400 KB SRAM; 4 MB Flash	Microcontroller: reads sensor, transmits over WebSocket	1250
MPU6050 Module	3-axis accel $\pm 2/4/8/16g$, 3-axis gyro $\pm 250\text{--}2000$ dps, I2C	Inertial sensing at 100 Hz	200
Li-Po Battery	3.7V, 500 mAh, rechargeable	Portable power supply	300
TP4056 Module	Li-Po charge controller with protection	Safe USB-C charging	40
3D-Printed Enclosure	PLA, wristband-integrated	Wearable form factor	120
Misc. (wires, connectors)	—	Assembly	80
Grand Total	—	—	₹1990

4.2 Software Stack

Stage	Action	Location
1. Sensor acquisition	Read ax, ay, az, gx, gy, gz at 50 Hz	On-device (ESP32-C3)
2. Wi-Fi transmission	WebSocket stream to dashboard server	Local network
3. Noise removal	Low-pass Butterworth filter (cutoff 20 Hz)	Dashboard (Python/Flask)
4. Bandpass filtering	4–12 Hz pass-band for tremor isolation	Dashboard
5. FFT analysis	256-point FFT; extract dominant frequency, spectral power	Dashboard
6. Feature extraction	RMS, variance, peak-to-peak, spectral centroid per 2-second window	Dashboard
7. Deviation scoring	Compare features against personal baseline z-scores	Dashboard
8. Neural network inference	1D-CNN + LSTM classifies event into 3 classes	Dashboard

Stage	Action	Location
9. Dashboard display	Plot raw signal, FFT, detected class, event history	Browser UI
10. User confirmation	Patient selects Yes/No for each flagged event	Browser UI
11. Baseline update	Confirmed event stored; model retrained incrementally	Dashboard backend

4.3 Phase 1 - Calibration Mode (Days 1–7)

During the first seven days of use, the system operates in Calibration Mode. The dashboard toggle is set to “Calibrative Mode”. The following sequence executes continuously: 1. Sensor streams 6-axis data to the dashboard. 2. The DSP pipeline applies noise removal and bandpass filtering. 3. Feature extraction runs on each 2-second window. 4. If the feature vector exceeds initial threshold rules (frequency in 4–6 Hz band, RMS above ambient mean), the system flags the event. 5. The dashboard immediately prompts: “Was this a tremor? [YES] [NO]” 6. The patient’s response is stored alongside the raw window and computed features. 7. After 7 days, the accumulated confirmed events are used to:

- (a) compute the personal baseline statistics and
- (b) train the TCN classifier. This phase generates the validated labelled dataset that no prior system has been able to collect at zero cost to clinicians or patients.

4.4 Phase 2 - Inference Mode (Day 8 onward)

The dashboard toggle switches to “Inference Mode”. The trained τ_{CN} now drives all event classification: 8. The TCN model evaluates each 2-second IMU window.

1. The model outputs one of three class labels: Thumb-Tapping Tremor, Driving Vibration, or Consolidated Random.
2. If the class is Thumb-Tapping Tremor, the dashboard shows a real-time alert with the detected waveform and frequency spectrum.
3. The system asks: “Was this a tremor? YES / NO ” to continue collecting ground truth for model refinement.
4. All confirmed events are logged to the longitudinal record and the model is retrained incrementally overnight.

This creates a self-improving system: as the patient uses it, it becomes more accurate for that specific individual. The shift from reactive clinical assessment to proactive, continuous, personalised monitoring is the central clinical contribution of this work.

5. Real-Time Dashboard

The dashboard is the patient’s primary interface. It runs in any modern browser connected to the same local Wi-Fi network as the wearable device. The interface is divided into four regions:

Dashboard Region	Contents
Mode toggle (top bar)	Calibrative Mode $\leftrightarrow$ Inference Mode toggle; current session timer; battery indicator
Live signal panel	Real-time 6-axis accelerometer and gyroscope trace at 50 Hz; scrolling 10-second window
FFT spectrum panel	256-point FFT updating every 500 ms; dominant frequency highlighted; 4–6 Hz tremor band shaded
Event log & test case panel	Chronological list of detected events labelled as Thumb-Tapping Tremor / Driving Vibration / Consolidated Random; each entry shows timestamp, dominant frequency, and patient’s Yes/No confirmation

6. Novelty & Innovation

This project is differentiated from all prior work by three interlocking innovations that, when combined, form a system no existing commercial or academic product replicates.

6.1 User-Confirmed Validated Labelling

Most wearable tremor systems generate an algorithmic output and treat it as ground truth. This is methodologically flawed: the algorithm cannot distinguish a Parkinsonian rest tremor from a bus vibration, a typing motion, or intentional finger tapping. This system instead flags candidate tremor events and immediately presents the patient with a simple binary prompt:

“Was this a tremor? YES / NO ”

Each confirmed or denied event is stored with its raw IMU waveform, FFT feature vector, and timestamp. This produces the first continuously growing, patient-confirmed labelled dataset for that individual - something no clinical trial has been able to generate at scale because the infrastructure did not previously exist affordably.

6.2 Person-Specific Baseline Modelling

Generic population-level models fail because every Parkinson's patient moves differently. Their tremor frequency, dominant axis, amplitude envelope, and rest-versus-action character vary enormously between individuals and even between the on-medication and off-medication states of the same individual. This system solves the problem by comparing a patient only against themselves.

During the seven day Calibration Phase, the system records the patient's full movement repertoire across all activities of daily living.

Innovation	Description	Why It Matters
Patient-confirmed labelling loop	System asks the patient to validate each detected event, producing ground-truth labels in the home environment.	First affordable system to generate validated, personalized tremor training data outside a clinic.
Person-specific baseline	Severity is measured as deviation from the individual patient's own normal, not a population average.	Eliminates false positives caused by inter-patient variability in natural movement patterns.
Dual-phase calibrate-then-infer	Explicit 7-day calibration window before autonomous alerting begins.	Prevents premature alerts; ensures the model is trained on the patient it will monitor.

24/7 continuous monitoring	Wearable captures full-day motion including medication on/off cycles, morning stiffness, and evening fatigue.	Clinical visits capture < 0.1% of the patient's actual daily motor behaviour.
₹800 total hardware cost	All components sourced domestically; no subscription or cloud dependency.	Accessible to patients in Tier-2 and Tier-3 Indian cities where neurology specialists are scarce.

7. Limitations

- Single IMU placement (wrist) does not capture full-body tremor. Foot tremor and jaw tremor, which are clinically significant in some patients, are not detected.
- The MPU-6050 exhibits known gyroscope thermal drift over long sessions. Drift correction is applied but residual low-frequency bias may affect baseline statistics over sessions longer than 4 hours.
- The TCN model requires a minimum of approximately minimum events per class before achieving reliable inference. Early Phase 2 performance may be lower than final performance.
- The system is not clinically validated. It does not diagnose Parkinson's disease and does not replace neurologist assessment. It is a monitoring and alerting tool only.
- Wi-Fi dependency means the system requires local network access. Offline edge inference on the ESP32-C3 is planned for a future iteration.

8. Future Scope

1. Add a second IMU at the ankle or fingertip to capture gait tremor and fine motor tremor simultaneously.
2. Making a smart Ring that helps detect small tremors in the initial phases of the disease.
3. Finding a severity score and display the exact time and date of tremor.
4. Replacing mpu6050 with a much better sensor.

9. References

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